Status	Finished
Started	Wednesday, 21 October 2026, 8:04 PM
Completed	Wednesday, 21 October 2026, 8:11 PM
Duration	7 mins 3 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search works on unsorted lists
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Linear search ✓
c. Binary search
d. Dynamic search

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. The last element is returned
c. An error is raised
d. The first element is returned

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It can be applied only if the list is sorted ✓
c. It compares elements sequentially
d. It works on unsorted lists

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search backtracks to find duplicate elements
c. The search starts over from the beginning
d. The search stops immediately ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search requires the list to be sorted.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search divides the list into halves.

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search stops
c. The search continues in the left or right sublist ✓
d. The list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(1)
- b. O👎
- c. O(log n) ✓
- d. O(n log n)

Question **9**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. log n
- c. n ✓
- d. n/2

Question **10**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and unsorted
- b. When the list is small and unsorted ✓
- c. When the list is small and sorted
- d. When the list is large and sorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O👎 ✓
- c. O(log n)
- d. O(n log n)

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Finding
c. Searching ✓
d. Discovering

Question **13**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Linear search ✓
c. Interpolation search
d. Binary search

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Sorting the list before searching
c. Checking each element sequentially ✓
d. Dividing the list in half repeatedly

Question **15**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search continues in the right sublist
c. The search ends successfully ✓
d. The list is sorted